

Abdulaleem Abdullah Ali Al-Patool

Software Developer & AI Engineer (Computer Science Student)

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PROFESSIONAL SUMMARY

Expected 2028 graduate seeking entry-level or internship roles in software development, backend engineering, or artificial intelligence (AI) and machine learning (ML) engineering. Hands-on experience with React and Python applications, FastAPI, Representational State Transfer (REST) APIs, ASP.NET framework, JSON Web Token (JWT) authentication, and PySide6 desktop applications. Practical Arabic Natural Language Processing (NLP) background: fine-tuned AraT5 on 250,759 Arabic news texts, reaching 97.40% main-category accuracy.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, Structured Query Language (SQL)

Frontend: React.js, Vite, HTML5, CSS3, TanStack Query, Responsive Web Design, Component-Based Development

Backend & APIs: FastAPI, REST APIs, JWT Authentication, Uvicorn

AI/ML & NLP: Machine Learning, Deep Learning, NLP, Arabic NLP, Transformers, Hugging Face, Sequence-to-Sequence Models, Fine-Tuning, Text Classification, Hierarchical Classification, Term Frequency-Inverse Document Frequency (TF-IDF), Naive Bayes, Feature Extraction, Feature Selection, Data Preprocessing, Dataset Preparation, Data Analysis, Confusion Matrix, Cross-Validation

Desktop & Tools: PySide6 (Qt for Python), Git, GitHub, Vercel, VS Code, JupyterLab, Google Colab, npm, Linux Command Line, Windows Command Line

Engineering Concepts: Data Structures, Algorithms, Algorithm Analysis, Sorting, Searching, String Matching, Numerical Methods, Software Architecture, Layered Architecture, Domain/Application/Infrastructure Separation, Requirements Engineering, Software Requirements Specification (SRS)

PROJECTS

Arabic News Hierarchical Classifier

Python, Hugging Face Transformers, AraT5, Arabic NLP, Google Colab

- Fine-tuned UBC-NLP/AraT5-base, a text-to-text Transformer, to jointly predict main and sub-categories of Arabic news.
- Prepared a dataset of 250,759 Arabic news texts from 7 Arabic news websites, consolidating categories into 6 main classes, including Economy News, Political, Science & Health, and Sports.
- Achieved 97.40% main-category accuracy, 96.58% accuracy at both hierarchy levels, 95.70% exact-match rate, and 0.0221 evaluation loss.

FWallet – Merchant E-Wallet Frontend (forked from github.com/ahmed-altaweel/FWallet)

React.js, JavaScript, Vite, TanStack Query, JWT, REST APIs | github.com/Abdulaleem-Al-patool/FWallet

- Developed the frontend for merchant e-wallet account management, implementing a User > Provider > Accounts data model.
- Wrote system requirements following an SRS based on ISO/IEC/IEEE 29148:2018 and translated them into a component-based React architecture.
- Integrated REST APIs with JWT authentication; used TanStack Query for server-state management, caching, and synchronization.
- Built reusable user interface (UI) components and a responsive interface for merchant account workflows.

Numerical Analysis Toolkit

Python, PySide6, Numerical Analysis

- Developed an interactive desktop application visualizing 8 numerical methods: Bisection, Newton-Raphson, Secant, Lagrange Interpolation, Cramer's Rule, Gaussian Elimination, Jacobi Method, Trapezoidal Rule.
- Implemented iteration-by-iteration visualization with calculation steps, error analysis, result tables, and graphs.
- Designed an interface where users enter parameters and observe how each method converges.

Personal Engineering Portfolio

React.js, JavaScript, Vite, CSS, Vercel

- Designed and developed a responsive, component-based portfolio site presenting software and AI/NLP projects, technical competencies, and contact information.
- Deployed on Vercel with a continuous deployment workflow from GitHub.

EDUCATION

Ibb University, Yemen

2023–2028 (Expected)

Computer Science & Information Technology

Relevant coursework: Databases, Algorithms, Numerical Analysis, Machine Learning, Deep Learning, NLP

LANGUAGES

Arabic – Native | English – Intermediate